



BARISTA
TRAINING
MANUAL

ATOMIC COFFEE ROASTERS



Continue your Coffee Education here

ABOUT US

Atomic's roots go back to the mid-90s, when two brothers opened a series of cafes on Boston's North Shore. What started as a local coffee vision has grown into Atomic Coffee Roasters—now led by the second generation of our family.

We combine decades of experience with the energy of a young, passionate team that loves to innovate and collaborate.



After 25+ years in coffee, we've seen the industry evolve, and we're here to help you thrive in it. Our wholesale team will guide you through everything from choosing the right coffee and equipment to staying ahead of trends and service styles.

TOOLS OF THE TRADE

A. HOPPER

where the beans are stored/loaded before grinding

B. BEAN GATE

when open allows beans to fall through, when closed prevents any beans from being ground

C. GRIND ADJUSTMENT COLLAR

to make your espresso grind coarser or finer

D. DIGITAL CONTROL SCREEN

controls how much coffee the grinder dispenses with each dose

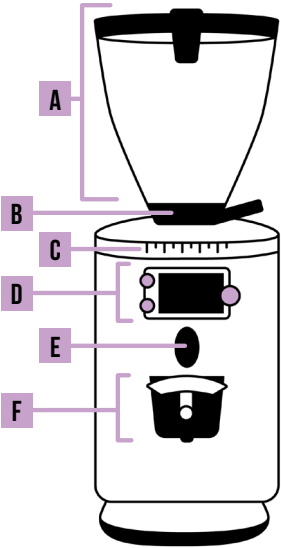
E. CHUTE

where the ground espresso comes out

F. PORTAFILTER BRACKET

cradles the head of the portafilter as you grind your dose into it

ESPRESSO GRINDER



ACCESSORIES



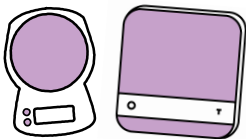
TAMP

to press and compact coffee grounds



KNOCK BOX

for knocking used espresso “pucks” out of your portafilter



SCALES

for best accuracy in weighing ground coffee and espresso

The espresso machine is the heart of the cafe—pumping water at 9 bars of pressure through finely ground coffee pucks to extract the perfect shot.

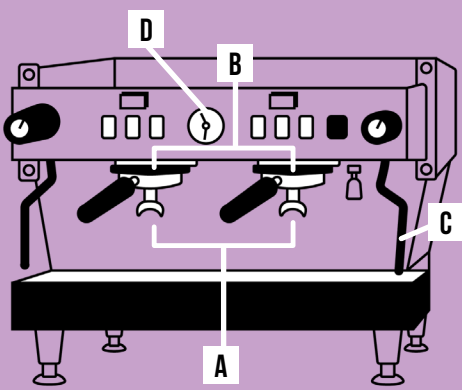
A. PORTAFILTER
where you dose and pack the coffee grounds

B. GROUPHEAD
where you lock in the portafilter

C. STEAM WAND
for heating, texturizing, and steaming milk

D. PRESSURE GAUGE
displays the pressure of your coffee/steam boiler, usually most important for troubleshooting purposes

ESPRESSO MACHINE



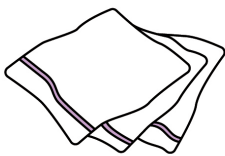
ACCESSORIES



MILK PITCHERS
to steam & texturize milk



SHOT GLASSES
to catch your shots & assess espresso quality



BAR TOWELS x3
1 for wiping steam wands, 1 to clean & dry portafilters & 1 for wiping messes



PULLING A SHOT

1 Remove portafilter and flush the grouphead

Pre-heating the group, and rinsing any leftover coffee particles.

2 Wipe the portafilter Make sure it's clean and dry.

3 Grind coffee dose into portafilter Try to have grounds fall in an even mound in the center of the portafilter.

4 Weigh dose on scale to confirm correct amount of coffee Essential for maintaining a proper recipe and coffee-to-water ratio.

5 Level/distribute Disperse and settle the bed of grounds to ensure even extraction.

6 Tamp Compact the espresso grounds, making sure to stay level.

7 Check your puck surface If there are any cracks or big divots, knock out the coffee and start again.

8 Load portafilter Gently re-insert the portafilter into its grouphead. Roughly slamming the portafilter back into the machine will create cracks and channels in the bed of coffee that will yield a poor extraction.

9 Pull shots/begin extraction Pull them into the rinsed shot glasses, or a mug.

10 Monitor/diagnose shots Proper timing and use of visual cues are essential to maintaining quality.

11 Finish and reset Knock out the puck and flush the grouphead.

UNDERSTANDING EXTRACTION VS STRENGTH

WHAT IS EXTRACTION? WHAT IS STRENGTH?

EXTRACTION

is the act of pulling soluble compounds out of coffee grounds by using water.

STRENGTH

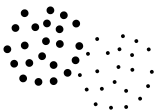
is the amount of solubles present in the finished cup, also called Total Dissolved Solids (TDS).

We control the extraction by manipulating variables that impact the water's access to soluble material, creating and affecting coffee flavor.



RATIO

This is the overall brew strength of your shot. How much coffee is going into the portafilter in relation to how much water we are running through it.



GRIND PARTICLE SIZE

The coarseness or fineness of your coffee. This directly relates to the rate at which the soluble material extracts from the coffee.

Finer grind = slower shots
Coarser grind = faster shots



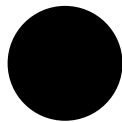
BREW TIME

How long we are allowing the water to extract soluble material from the puck of coffee.

In order to make consistently delicious espresso shots, it is extremely helpful to use a recipe, or Ratio — this will keep your **STRENGTH** constant so that you can focus on manipulating the **EXTRACTION**.

We will use scales to weigh the amount of ground coffee going into the portafilter, and to weigh the amount of liquid espresso coming out of the grouphead. A popular ratio that complements a lot of coffee is 1:2, or 1 part ground coffee IN, to 2 parts espresso OUT. Depending on the size of your portafilter, this might be something like 18 grams of ground coffee IN, to 36 grams of espresso OUT.

QUICK HITS



GROUND COFFEE



ESPRESSO

Example: 18g ground coffee | 36g espresso

DIALING IN AND ADJUSTING YOUR GRIND SETTING

- When a barista says they are 'dialing in,' it means they are making adjustments to their espresso grinder, in order to make the shots extract better.
- First, verify your dose weight (Ratio) is what you intended. If shots are running too slow, (generally reaching their total weight 6-10 seconds slower than your intended recipe) that is referred to as **over-extracted**.
- If your shots are running too fast, (generally reaching their total weight 6-10 seconds faster than your intended recipe) this is referred to as **under-extracted**.
- Adjust your grind size coarser or finer to correct the problem. Purge one portafilter of coffee after each grind adjustment to clear the chamber of any grounds from the previous setting.
- Now you are ready to pull your next shot, to confirm if you have brought your espresso back to an ideal extraction!
- Repeat the grind adjustment as needed, until you are **DIALED IN!**

OVER-EXTRACTED

The water has pulled too much soluble material from the bed of coffee. These shots often taste bitter, with a lingering unpleasant finish.

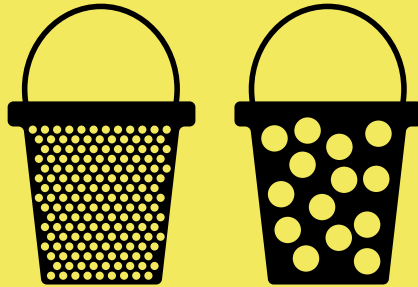
UNDER-EXTRACTED

The water has not spent enough time with the bed of coffee and has not dissolved enough soluble material. These shots often have a thin mouthfeel, with a sour or papery taste.

UNDERSTANDING GRIND SETTINGS

If you have a hard time recalling which way to adjust your grinder when dialing in, use this example:

IMAGINE YOU HAVE TWO BEACH BUCKETS.



SAND

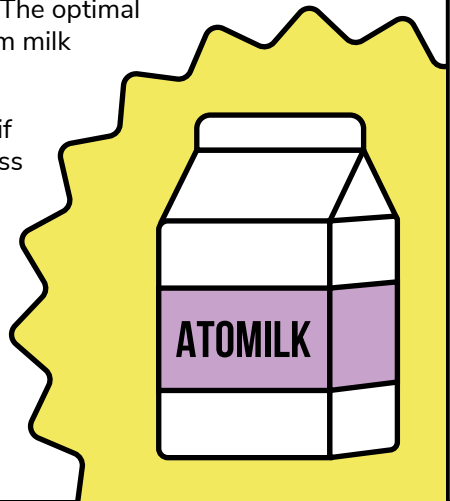
PEBBLES

One is packed full of beach sand, and the other is packed full of pebbles. If you were to pour water over the top of the two buckets, which one would the water flow through faster?

The sand is like your fine grind, and the pebbles are like your coarse grind.

EVERYTHING YOU NEVER THOUGHT YOU'D NEED TO KNOW ABOUT MILK

- Milk can be a great companion to espresso because the sugars and fats complement the bitterness and dial down the intensity.
- Whole milk is the best for steaming practice, and it is actually only about 4% milkfat. The easiest non-dairy for steaming is usually a “Barista” version of Oat Milk.
- Properly steamed milk will be naturally sweet with a smooth, even texture. The bubbles should be almost invisible to the eye; this is called “microfoam”.
- It should have a glossy appearance and viscosity similar to a can of white house paint.
- Dairy milk burns at 165°F. The optimal temperature range to steam milk within is 140°F-150°F.
- Latte art is **ONLY** possible if you’ve mastered the process of steaming milk.



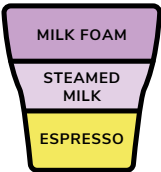
COMMON ESPRESSO DRINKS



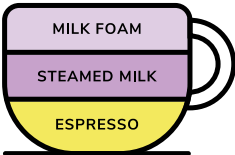
ESPRESSO
2oz



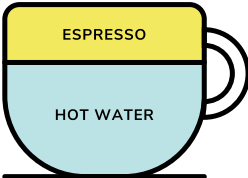
MACCHIATO
3oz



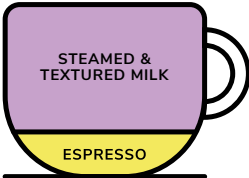
CORTADO
4oz



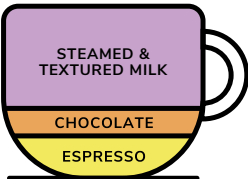
CAPPUCCINO
8oz



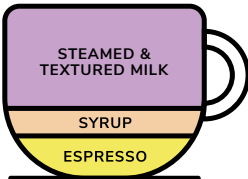
AMERICANO
12oz



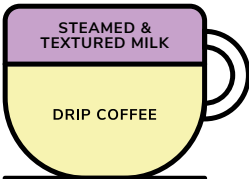
LATTE
12oz



MOCHA
12oz



FLAVORED LATTE
12oz



CAFE AU LAIT
12oz



HOW TO STEAM MILK

STEP-BY-STEP

- 1** Pour milk into a pitcher, about 1/4 inch below the spout.
- 2** Purge the steam wand, and place it into position — imagine it as aiming toward your toes.
- 3** Submerge the tip of the steam wand into the center of the milk, just barely covering the end.
- 4** Hold the pitcher handle with one hand. The other hand will activate the steam wand, and then immediately come down to rest against the side of the pitcher to monitor the increase in temperature.
- 5** When you're ready, turn on the steam wand and immediately begin the stretching/aerating phase. It should sound like a soft hissing or paper-tearing noise.
- 6** As soon as the milk's temperature becomes 100°F, raise the pitcher up about 1/4 inch and give it a slight tilt.
- 7** At this point the hissing noise should cease and you should see a circular motion, or "vortex" begin to happen.
- 8** When the pitcher becomes too hot to comfortably touch with your free hand, reach up and turn the steam wand off.
- 9** Lower the pitcher and immediately wipe the milk residue off your steam wand, with a downward twisting motion.
- 10** Give the pitcher a light tap against the counter to knock out any large bubbles. Then swirl the pitcher to mix the milk. You should see it turn glossy, looking like "wet paint" or "melted ice cream"
- 11** Optional: Use an instant read thermometer to confirm the temperature is between 140-150 degrees.



HOW TO POUR LATTE ART

STEP-BY-STEP

1 Pour your espresso into the cup and cradle it in the palm of your hand, tilting it towards the pitcher in your other hand.

2 Start your pour up high (about 3-4" above the surface of the espresso) to create the canvas for your art. Ideally the milk should dive below the surface of the espresso, preserving the rich brown color as much as possible.

3 When the cup is about 2/3 full, quickly bring the tip of your pitcher down and centered, as close to the liquid in the cup as possible. At this point you should see a crisp white dot begin to appear, because the milk is gliding across the surface (if you do not see white appear, you are not close enough).

4 Once you have formed the dot, hold steady and it will continue to grow as your cup fills.

5 The final step is called the "pull through." Once the cup is full, lift your pitcher back to the starting position and pull it directly across the center of your dot. When done correctly, this will appear as if an invisible thread is pulling through your dot, turning it into a heart. The pull through movement should mimic a helicopter taking flight, rather than an airplane.

**REMEMBER TO PRACTICE,
AND DON'T BE TOO HARD
ON YOURSELF— LATTE
ART IS FUN TO MASTER,
BUT IT DOESN'T TASTE
LIKE ANYTHING!**

WORKFLOW & EFFICIENCY

WHETHER WORKING SOLO OR WITH A TEAM, BAR
FLOW SHOULD BE CLEAN, CONSISTENT, AND FAST.

SPEED MUST NEVER COMPROMISE QUALITY.



WORKING SOLO

- Prep empty cup and pour cold milk into pitcher
- Pack and pull shot, then immediately begin to steam, so that espresso and milk finish at the same time.
- Pour milk, hand off drink, and clean.

WORKING WITH A PARTNER/ DOUBLE BARRING

For when you start to experience a backlog or build up of drinks in your queue.

- Main barista can request assistance, and should delegate their most pressing need to the secondary barista who slides over to help.
- To work together efficiently, the secondary barista should begin setting up cups and milk pitchers for each drink in the queue
- Main barista will pull shots and both people can steam if the round of drinks requires
- Communicate clearly about drink order and timing.

BEST PRACTICES

- Use separate rags for portafilters, counters, and steam wands.
- Clean as you go—wipe surfaces, flush machines, and empty pucks.
- Group drinks to maximize your machine's efficiency, e.g. on a 2 Grouphead machine, you can make an Americano, Latte, and Chai Latte all in one round.

ICED DRINKS

ICED ESPRESSO DRINKS

**COLD DRINKS REQUIRE SLIGHT
ADJUSTMENTS IN ASSEMBLY TO PRESERVE
BALANCE AND AVOID DILUTION.**

ICED LATTE WORKFLOW

- 1** Fill a cup with ice.
- 2** Add milk to the bottom of the cup (leave room for espresso).
- 3** Pull shots and pour them directly over the top.
- 4** Stir or use a tumbler to mix.



NOTE

Always pull espresso fresh to order. Never pre-pull and store for iced drinks.

BATCH BREW

BEST PRACTICES

- Use **freshly ground coffee**—grind + brew should be thought of as 1 married step.
- Brew with recipes **dialed in** for your machine.
- Never interrupt a brew cycle to pour early—wait until it's finished.
- **Discard** any coffee that has been sitting for over 60 minutes.
- **Label** urns or airpots with the brew time and start a timer to track freshness.



EQUIPMENT CLEANING

THERE IS NOTHING WORSE THAN HAVING YOUR EFFORTS THWARTED BY DIRTY EQUIPMENT.

**CLEANER EQUIPMENT
=
BETTER TASTING COFFEE**

**HAVE THESE
ITEMS HANDY
TO MAKE
CLEANING
EASIER:**

- Pallo tool
- Stubby flathead screwdriver
- Cafiza
- Six-quart bucket
- Blind backflush disc

EQUIPMENT CLEANING

ESPRESSO MACHINE (DAILY)

- 1** Remove portafilter baskets and screens.
- 2** Soak all metal parts (portafilters, baskets, screens) in a container with hot water and Cafiza.
- 3** Scrub grouphead gaskets and dispersion screens with the Pallo tool.
- 4** Backflush each group-head:
 - Insert a blind basket with 1/2 tsp Cafiza.
 - Activate the grouphead in 5 second bursts, pausing between each.
 - Repeat 5 times, then rinse and repeat with plain water to clear detergent.
- 5** Reinstall all parts and run a final water flush.
- 6** Wipe down the drip tray, body panels, and counter area.
- 7** Polish the machine and ensure the area is clean before closing.

EQUIPMENT CLEANING

GRINDERS (DAILY)

- 1 Close the hopper chute and grind out any remaining coffee.
- 2 Power off the grinder.
- 3 Use a grinder brush to remove grounds from burrs, chute, and fork.
- 4 Wipe exterior with a damp cloth.

BREWERS & SERVERS (DAILY)

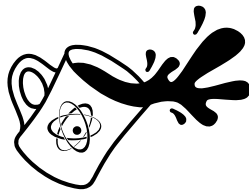
- 1 Dump any remaining brewed coffee.
- 2 Run a cleaning cycle using 1-2 tsp of Cafiza (no filter).
- 3 Scrub brew baskets and urn interiors with a long-handled brush.
- 4 Rinse thoroughly with hot water.
- 5 Wipe down outer surfaces and leave components to air dry.

Note: Use designated brushes and cleaning cloths for each area. Keep chemical containers and cleaning tools stored neatly after use.

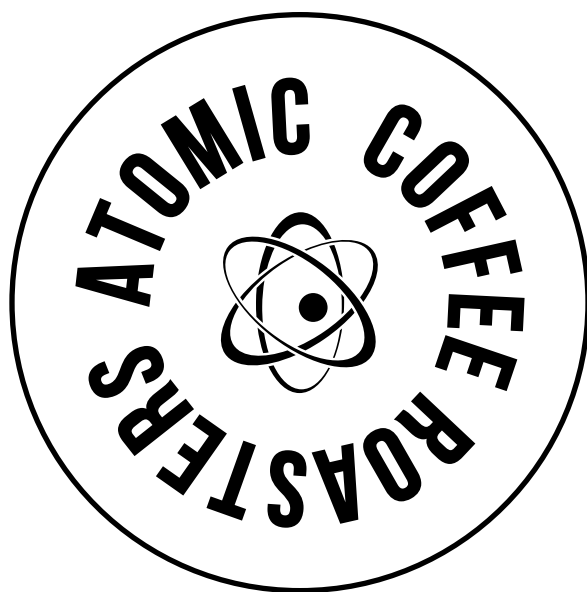
ATOMIC COFFEE ROASTERS

**THIS MANUAL IS DESIGNED TO HELP ATOMIC COFFEE
ROASTERS' BARISTAS AND PARTNERS DELIVER
CONSISTENT, HIGH-QUALITY BEVERAGES.**

**WITH ATTENTION TO DETAIL, REGULAR PRACTICE, AND
CARE, ANYONE CAN MASTER THESE SKILLS AND SERVE
COFFEE THAT DELIGHTS.**



**REACH OUT TO YOUR
ATOMIC REP WITH
ANY QUESTIONS!**

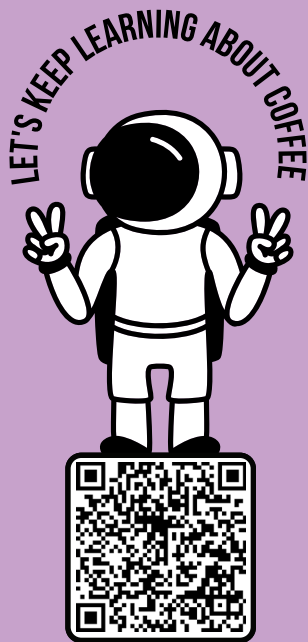


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